

A Zero Decoding Approach to Video Classification

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Abstract—Classifying videos into distinct categories, such as Sport and Music Video, is crucial for multimedia understanding and retrieval, especially with growing content volume. Traditional methods require video decompression to extract pixel-level features like color, texture, and motion, thereby increasing computational and storage demands. We present a novel approach that examines only the compressed bitstream of a video to perform classification, eliminating the need for bitstream decoding. To validate our approach, we built a comprehensive data set comprising over 29,000 YouTube video clips, totaling 6,000 hours and spanning 11 distinct categories. Our evaluations indicate precision, accuracy, and recall rates consistently above 80%, many exceeding 90%, and some reaching 99%. The algorithm operates approximately 15,000 times faster than real-time for 30fps videos, outperforming traditional Dynamic Time Warping (DTW) algorithm by seven orders of magnitude and state-of-the-art video classification model by three orders of magnitude.

Index Terms—video classification, bitstream analysis

I. INTRODUCTION

Video classification is fundamental for multimedia services, enabling functionalities such as content retrieval, recommendation, and optimized coding. Traditional techniques focus on analyzing video features such as color, texture, and motion, while recent approaches incorporate deep learning-based algorithms [1] [2] [3].

Current approaches follow a *pixel to understanding* paradigm, which depend on pixel-domain features and necessitate full decoding, leading to a storage increase ratio of up to 75:1 for a 1080p30 video compressed at 10 Mbps, as well as computational inefficiency on such massive data. This issue is exacerbated by high-volume video uploads to platforms like YouTube and TikTok, where videos are typically compressed. These methods are inadequate for classifying the 30,000 hours of video uploaded to YouTube every hour, even with optimizations such as down-sampling spatial and temporal resolutions and specialized low-memory-footprint classification techniques [4] [5]. Furthermore, these techniques often struggle with low-quality videos and raise privacy issues as decryption is needed. In some cases, DRM policies prevent decryption during transmission.

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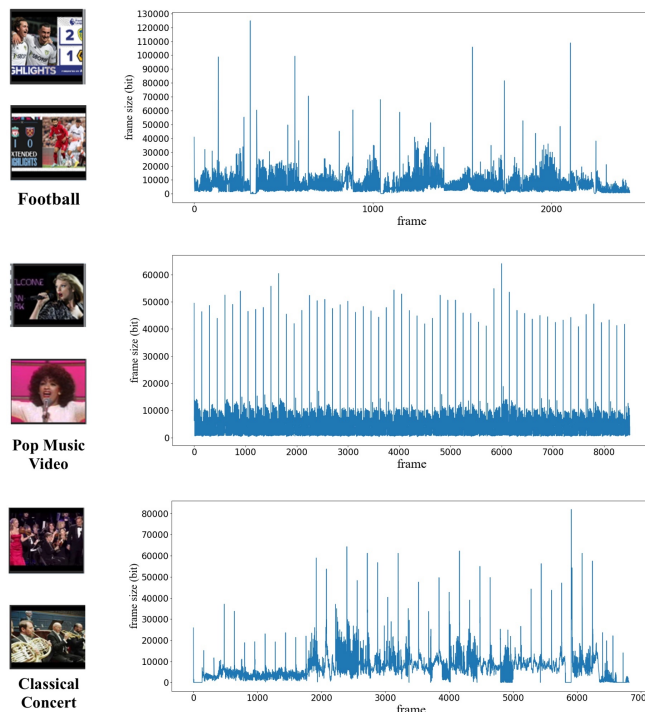


Fig. 1. Frame sizes (in bits) after encoding, over a timeline of frame numbers. Different categories exhibit unique patterns in frame size sequences.

To overcome these limitations, we adopt a *bitstream to understanding* paradigm, deriving an abstract video representation directly from compressed video bitstreams. We observe that sequences of video frame sizes extracted from compressed bitstreams provide highly compressed yet sufficiently rich representations of spatial and temporal characteristics by leveraging the advanced spatial and temporal prediction methods in modern video coding standards such as H.264/AVC, H.265/HEVC and H.266/VVC. Optimized encoders also incorporate advanced rate control algorithms, which allocated enforce bit rates across and within frames.

At a high level, the complexity of individual frames determines the size of the reference frames; uniformity of the frames determines the size of the predicted frames; the frequency of scene changes is determined by factors such as camera movement rhythms. As illustrated in Figure 1, encoding creates distinct patterns of encoded frame size sequences for

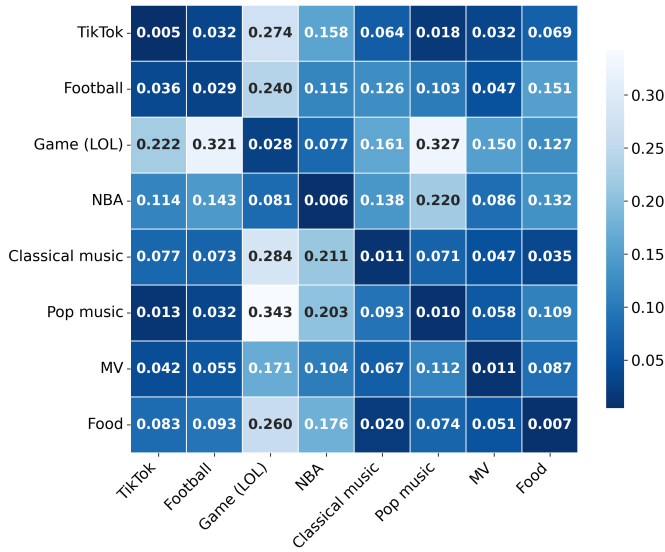


Fig. 2. KL divergence between videos from broad categories.

videos from different categories. This concept is generalized in Figure 2, where relative entropy values are multiples higher for videos from different categories.

Building on these insights, we propose a novel direction for video classification that represents videos as sequences of compressed video frame sizes, eliminating the need for bitstream decoding or parsing. We use these sequences as input to a ResNet-based deep neural network and demonstrate the capability to classify videos across both broad YouTube categories and across individual influencers.

Our approach operates thousands of times faster than real-time while consuming hundreds of times less storage than traditional approaches. It consistently maintains precision, accuracy, and recall rates of at least 80%, often exceeding 90%, and within a few percentage points of state-of-the-art video classification while ensuring privacy.

II. BACKGROUND

Video recognition has been extensively studied using pixel-domain features, including multimodal approaches [6], object-level understandings with Convolutional Neural Nets (CNNs) [7] [8], and finer-grain tasks involving recognition of intent and actions [9] [10]. Coarse-grain video-level classification has seen limited progress since the 1st Youtube-8M challenge, where WILLOW's Gated NetVLAD, a CNN based ensemble, set a benchmark for speed and performance [11] [12].

Studies have also explored compressed domain features for computer vision tasks. Notable works include substituting optical flow with motion vectors [13], using CNN with motion vectors and selective textures [14], and incorporating features such as I-frames, residuals, and audio waveforms [15]. These methods, while more efficient than full decoding, still require entropy decoding. Our method avoids any decoding by representing videos as sequences of video frame sizes, obtained directly from the compressed bitstream.

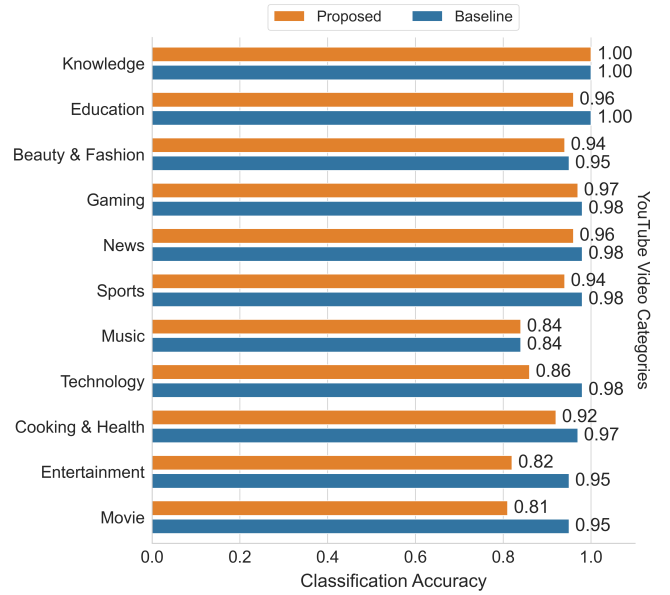


Fig. 3. Classification effectiveness on broad YouTube categories: WILLOW (baseline) vs proposed method.

Our approach reframes the problem of video classification as a typical Time Series Classification (TSC) problem [16]. We define the sizes of compressed frames in bits, in encoding order, as a time series $X = [x_1, x_2, x_3, \dots, x_T] \in \mathbb{R}^T$, where T represents the length of the time series. Consequently, video classification can be formulated as a mapping between X and a one-hot label vector Y .

Traditional TSC algorithms often use distance-based metrics with a k -nearest neighbor classifier (K -NN). Such algorithms include Dynamic Time Warping (DTW) [17], Weighted DTW [18], Move-split-merge [19], Complexity-Invariant Distance [20], Derivative DTW [21], Derivative Transform Distance [22], Elastic Ensemble [23], etc., where DTW is often used as a baseline [16]. All such methods take the entire time series as input and are computationally intensive. Even following the reduced complexity TSC algorithm [24], TSC algorithms are in general, still highly time-consuming. In recent years, deep learning based TSC has been studied broadly. A study by Fawaz et al. [25] evaluated the performances of nine deep learning models for TSC on the univariate UCR/UEA archive benchmark [26] and 12 Multivariate Time Series (MTS) data sets [27]. Among them, the most recent and best performing are Encoder [28] and Residual Network (ResNet) [29], respectively.

Based on the above studies, we used WILLOW as a baseline for pixel-level video classification, DTW as a baseline for traditional TSC algorithms, and ResNet as the backbone for our time series classifier.

III. BITSTREAM TO UNDERSTANDING

A. Distinct Video Categories have Distinct Distributions

We find that videos across various categories exhibit unique style and editing traits, influencing the encoding decisions

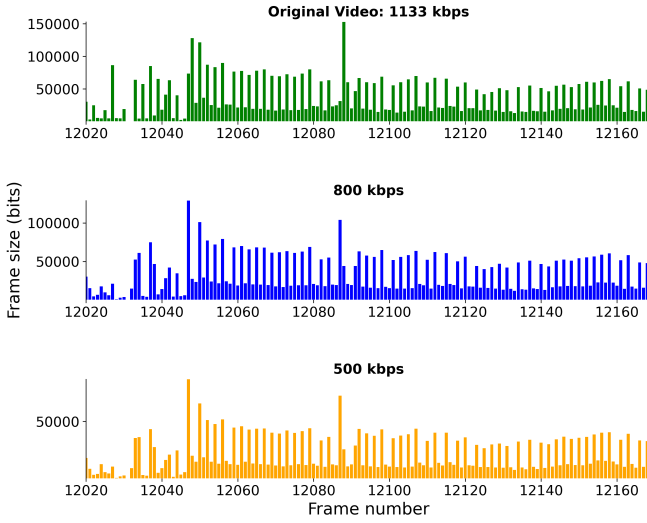


Fig. 4. Bitrate variation for the same clip transcoded to different bitrates.

made by optimized video encoders. The disparity in temporal bitrate patterns, as shown in Figure 1, may be attributed to factors such as scene complexity, object motion, camera motion, and edits, which are intrinsic to each genre. For instance, in football, sporadic high peaks correlates with fast-paced action and rapid scene changes. Videos from news and classical concert segments, in contrast, depict lower variability in bitrate, due to predominantly static backgrounds and slower-paced visual changes. Similarly, pop music videos, with their dynamic lighting and camera movements, show more frequent spikes than classical concerts, which typically contain steady camera shots, limited variation in composition, and uniformity in lighting conditions from frame to frame. These resulting encoded bitstreams provide revealing high-level statistics, useful for classifying videos with pronounced stylistic differences, such as NBA vs. Football or Pop vs. Classical concerts.

We also find that content from social media influencers tend towards more volatile bitrate distributions compared to broadcasts television, suggesting less uniformity in video composition and wider variety of shot types. This is likely due to the diverse nature of such content which do not adhere to a particular production style. This volatility suggest the need for either a sufficiently complex model, or sufficiently long time series, in order to discern subtle nuances between content that might otherwise be similar thematically.

B. Correlation between Transcoded Bitstreams

As shown in Figure 4, a test clip originally downloaded at 1.13Mbps was transcoded to both 1Mbps and 500kbps. While the encoding setting differed from the original YouTube version, the frame size time series of various bit-streams remained highly correlated.

In our experiment, we standardized all clips to 1.5Mbps bitrate using FFmpeg’s H.264/AVC encoder. We then calculated the inter- and intra-class Kullback-Leibler Divergence (KLD) for 8 sub-classes within our data set, namely NBA, TikTok, Football, Classical Concerts, Pop Concerts, Gaming

(League of Legends), Music Videos, and Culinary Exploration. As Figure 2 illustrates, intra-class KLD values are typically orders of magnitude lower than inter-class KLD values.

These results indicates high correlation between the frame size time series of bitstreams with various encoding settings, suggesting that 1) post-compression frame sizes, reflecting bitrate variations, is useful in video classification; and 2) a single model might be capable of classifying videos across a spectrum of bitrates.

C. Model architecture

We follow [30] and opt for the established ResNet time-series classifier which mainly consisted of 3 residual convolutional blocks with 256, 512, 512 filters each for feature extraction. Predicting the class of a given input X can be expressed as:

$$\begin{aligned}\hat{y} &= f_L(\theta_L, x) \\ &= f_{L-1}(\theta_{L-1}, f_{L-2}(\theta_{L-2}, \dots, \\ &\quad f_1(\theta_1, x + \mathcal{F}(x)))\end{aligned}\quad (1)$$

where $\hat{y}$ is the predicted output, f_L is the composite function (convolution, batch normalization, activation) applied at the L -th layer, θ_L denotes the set of parameters at the L -th layer, and $\mathcal{F}(x)$ denotes the shortcut connection in each residual block. The final output $\hat{y}$ of the ResNet model is produced by a global average pooling layer followed by a softmax classifier. The global average pooling layer generates feature maps of $\hat{y}$ for different video categories in this classification task. The softmax classifier maps these feature maps to a probability distribution over the output classes. Residual connection between convolutional layers effectively avoids vanishing gradients in the training process [30].

D. Model Training Specifics

We used fixed parameter initialization and an auto-adjusted learning rate, with factor set at 0.5 and patience level at 40. To avoid overfitting, an early stop strategy was adopted with the patience set to 80. The adam optimizer was used for all contrast experiments.

For this multi-classification task, categorical cross-entropy loss function is used as the supervised training loss:

$$\text{Loss}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (2)$$

where y_i denotes one-hot encoding for the corresponding video category, $\hat{y}_i$ denotes ground truth, and C denotes the total number of categories.

IV. EXPERIMENTS

A. Data Preparation

Existing data sets, such as YouTube-8M [31], Activity Net [32], and Net Sports-1M [33], focus primarily on short-form content, often limiting clips to less than 5 seconds.

We curated a more comprehensive data set, using keyword search on Youtube to collect 29,142 video segments,

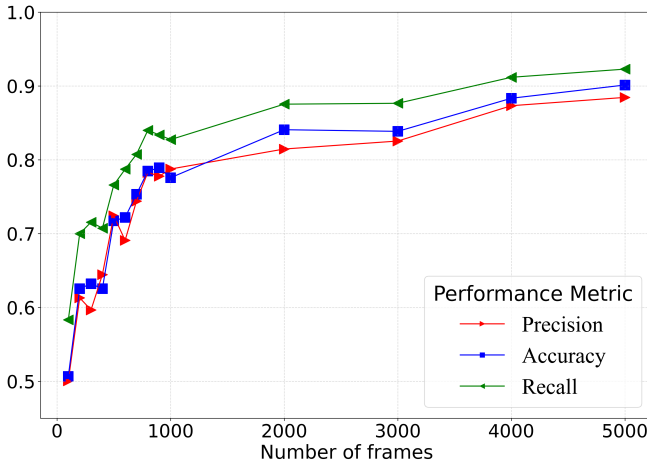


Fig. 5. Classification performance as a function of input size.

each containing a minimum of 3,000 frames, spanning 11 broad YouTube categories: Movies, Entertainment, Cooking & Health, Gaming, Technology, Knowledge, Music, Sports, Beauty & Fashion, News, and Education, collectively exceeding 6,000 hours in duration. Each category holds a minimum of 400 clips that vary in spatial resolution, from 298x480 for TikTok videos to 720p and 1080p for others, but are consistent in frame rate (either 30fps or 60fps) within their respective category.

Clips in some categories overlap conceptually (TikTok may contain NBA or Dance clips), testing the algorithm’s ability to discern “main characteristics” of the content, such as distinguishing a TikTok-style NBA clip from a standard NBA broadcast. Because we built the training and testing data sets using search and meta-data, we tend to only discover highly popular and viral clips for each category. For an NBA-themed clip to become viral on TikTok, it usually follows a certain recognizably TikTok style, as a result of the TikTok filters, editing tools and various social marketing “rules” which makes it distinctively different from a clip of standard NBA broadcast. We aim to distinguish an NBA clip as such when it originates from a live broadcast, but to identify that same clip as TikTok content once it has been integrated and disseminated within the TikTok platform. In no cases was the proposed classification algorithm in the loop of test clip class verification and assignment to the training and test data sets.

B. Classification Performance

We find that entropy-coded bitstreams demonstrate particular sensitivity to distinct “editing styles”, including shot selection, camera angles, and movement, which allows the classifier to distinguish diverse video categories such as NBA games, football matches, and classical or pop concerts, even when each video frame is represented by a singular numerical value. In high-level video categories (defined by YouTube), we observe precision, accuracy, and recall rates consistently exceeding 80%, as shown in Figure 3, with a slight decline in categories with greater style diversity. Moreover, our classifier

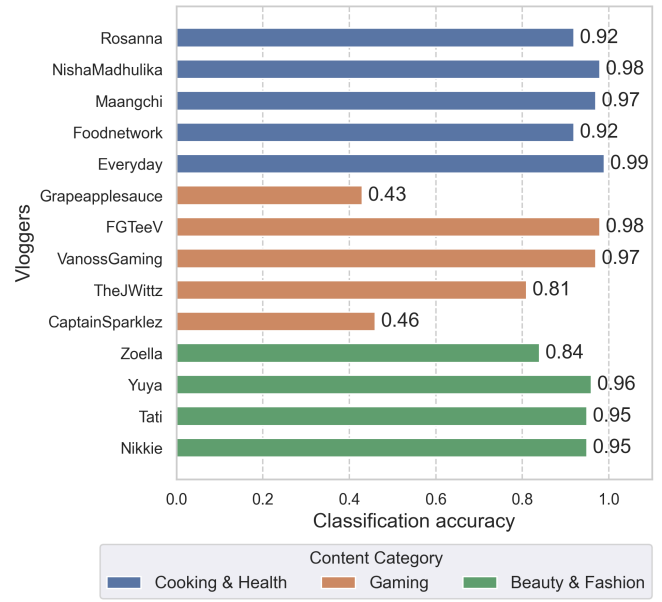


Fig. 6. Evaluating ability to discriminate individual vloggers.

achieves accuracy within five percentage points of state-of-the-art method (WILLOW) across broad categories, while being significantly simpler in architecture and requiring substantially less data. As shown in Table II, WILLOW uses 360 RGB frames as input while our proposed method only requires 3000 floating-point numbers, each representing the size of a frame.

For finer grain classification, we demonstrate that compressed frame sizes could capture distinctions between influencers within the same video category as illustrated in Figure 6. In Gaming, however, classification is ambiguous due to uniformity of video styles—typically a floating-head commentary overlaid on gameplay—resulting in similar motion patterns.

C. Classification Speed

Our ResNet-based video classifier, running on a single Nvidia A100 GPU, processed 1,818 test videos—each containing 3,000 frames—in less than 13 seconds, achieving a real-time factor of approximately 15,000 for 30fps, within striking distance of the 30,000-to-1 ratio of total video duration constantly uploaded to YouTube.

Conventional non-deep-learning-based Dynamic Time Warping (DTW) [17] requires an extensive 27 hours to classify short 30-frame clips from the same dataset, approximately 7.5×10^7 times slower than the ResNet classifier, with an accuracy of less than 0.35, making it an impractical alternative. Similarly, the deep-learning-based WILLOW model took over 40 seconds to classify a single 3000-frame video (Table II), over 10^4 times slower than our proposed method.

D. Robustness across coding parameters

We show robustness of our classifier across mainstream video encoders H.264 (AVC) and H.265 (HEVC), across encoding settings, and when classifying bitstreams encoded

TABLE I. Classification Performance at various train/test bitrates.

Metric (%)	Constant Bitrate (kbps)																Average Bitrate (kbps)				Training
	without B-frames (AVC)				with B-frames (AVC)				without B-frames (HEVC)				with B-frames (HEVC)				with B-frames (AVC)				Bitrate
	800	1000	1200	1500	800	1000	1200	1500	800	1000	1200	1500	800	1000	1200	1500	800	1000	1200	1500	(kbps)
Precision	84.49	80.99	70.50	56.61	85.11	79.66	67.16	58.17	84.61	84.48	84.22	82.34	85.98	85.93	84.82	82.87	86.87	85.35	79.39	70.15	800
Accuracy	83.79	74.01	46.16	20.67	85.27	74.63	42.20	21.16	84.14	84.22	83.52	81.72	85.74	85.68	84.45	82.54	87.62	83.54	72.40	52.35	
Recall	85.33	76.33	50.64	28.39	85.72	77.27	48.34	29.31	84.42	82.34	84.46	83.34	86.14	86.46	85.28	84.11	88.01	84.79	75.90	56.91	
Precision	78.63	83.47	80.89	68.80	85.16	87.93	84.80	66.10	83.21	84.42	84.62	83.66	85.12	85.84	84.78	84.17	86.82	87.98	85.55	78.12	1000
Accuracy	78.22	83.29	76.36	42.45	86.14	87.50	80.94	34.78	83.06	84.09	84.45	83.21	85.12	85.84	84.81	84.24	88.24	88.00	84.90	72.90	
Recall	78.47	84.44	78.62	48.08	86.67	88.14	82.67	42.65	82.82	84.10	85.09	83.86	85.81	86.16	85.50	84.92	88.65	88.48	86.00	76.65	
Precision	69.78	77.83	82.18	79.59	79.19	84.65	85.76	79.34	83.56	85.29	84.42	83.80	85.64	87.54	87.63	86.63	69.78	75.19	74.86	72.25	1200
Accuracy	64.11	75.99	81.19	74.38	80.94	85.40	86.26	73.76	83.16	85.02	84.09	83.32	85.53	87.59	87.38	86.56	56.06	70.92	71.53	63.00	
Recall	64.25	76.32	82.81	77.94	81.74	86.48	87.47	76.69	83.24	85.48	84.41	83.92	85.80	87.82	87.88	86.94	59.07	74.62	75.56	68.84	
Precision	59.41	72.15	78.54	84.11	66.07	73.66	81.19	84.55	81.77	83.64	84.43	84.77	83.08	85.56	86.21	87.60	72.67	78.55	84.76	85.23	1500
Accuracy	58.04	72.77	79.58	82.92	61.26	72.15	80.82	83.17	81.51	83.32	84.14	84.66	83.16	85.63	86.25	87.59	73.14	80.45	85.77	85.64	
Recall	55.06	72.01	80.28	84.41	60.65	71.06	80.70	83.24	81.62	83.57	84.85	84.78	83.98	85.89	86.68	87.93	73.68	80.56	86.30	86.35	

TABLE II. Comparison with state-of-the-art video classification model. Complexity is assessed as the total number of parameters. Accuracy and Runtime are averaged over the entire test set.

	Baseline (WILLOW)	Proposed
Input Size	$1920 \times 1080 \times 3 \times 360$	3000
Complexity	999,704,219	2,013,058
Runtime per Video	15s (decoding) + 25s (pixel-domain feature extraction) + 0.005s (inference)	no decoding + no pixel-domain feature extraction + 0.004s (inference)
Accuracy(%)	0.964	0.917

at bitrates different from those used during model training. Table I shows the classification performance across various combinations of training bitrates, testing bitrates, and encoding settings (AVC, HEVC, Average Bitrate (ABR) mode, Constant Bitrate (CBR) mode, with and without B-frames). For HEVC, model performance shows minimal degradation when classifying videos with mismatched bitrates. For AVC, classification remains effective for videos encoded at about 50% of model training bitrate, but performance suffers when the input video bitrate is reduced to 1/3 of the model's training bitrate.

E. Impact of Input Length

We assess the classifier's performance as influenced by the number N of frames utilized for model training and classification with a range of N values: 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 2000, 3000, 4000, 5000 frames. The corresponding classification outcomes for the Average Bitrate (ABR) scenario are depicted in Figure 5.

The results indicate that smaller N values may compromise classification performance, likely due to localized content variations. As N increases, classification efficacy improves until converging asymptotically to a stable state.

V. CONCLUSION

In this work, we demonstrate the feasibility of classifying videos without decoding, relying solely on examining their compressed video bitstreams. We implement a ResNet-based classifier that uses the time series of compressed video frame sizes as input. Our approach achieves robust classification capabilities, with a minimum of 80% accuracy, across 11 broad categories and 39 specific vloggers in 29,142 video segments and across various bitstream encoding settings (AVC, HEVC, ABR, CBR, B-frames).

This design *respects data privacy* as frame sizes are readily accessible through various means, such as byte-aligned headers or Network Abstraction Layer (NAL) packets without decoding. Remarkably, NAL packet-based analysis can even function with encrypted content, positioning our technique as a scalable solution for network carriers. Furthermore, this design *vastly reduces computational complexity*, allowing processing of 30fps videos at approximately 15,000x faster than real-time, outperforming traditional DTW algorithm by seven orders of magnitude and state-of-the-art pixel domain classification model by three orders of magnitude in both runtime per video and model complexity, with only a five percentage point reduction in overall accuracy.

This paper marks the first attempt at classifying video using compressed bitstreams. Further research include addressing the high number of frames ($N > 2000$) required for training to achieve accuracies above 80%, establishing the maximum number of classes this method can distinguish, and assessing its efficiency on other sequence models.

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